UNIVERSITY OF EDINBURGH

COLLEGE OF SCIENCE AND ENGINEERING

SCHOOL OF INFORMATICS

INFR11145 TEXT TECHNOLOGIES FOR DATA SCIENCE PG AND INFR11229 UG

Monday 6th May 2024

13:00 to 15:00

INSTRUCTIONS TO CANDIDATES

1. **Note that ALL QUESTIONS ARE COMPULSORY.**
2. **DIFFERENT QUESTIONS MAY HAVE DIFFERENT NUMBERS OF TOTAL MARKS. Take note of this in allocating time to questions.**
3. **This is a NOTES NOT PERMITTED, CALCULATORS PERMITTED examination.**

   Notes and other written or printed material MAY NOT BE CONSULTED during the examination.

   CALCULATORS MAY BE USED IN THIS EXAMINATION.

Year 4 Courses

Convener: K.Etessami
External Examiners: J.Bowles, A.Pocklington, R.Mullins

THIS EXAMINATION WILL BE MARKED ANONYMOUSLY

Please provide answers to **ALL** of the following questions. It is more important to have a clear answer than a longer answer. When calculations are needed, please **show steps** and give answer to **three** decimal digits.

1. (a) For each of the following preprocessing steps, explain the step, and list an advantage and a drawback with an example of each situation: [*3 marks*]

   i. Stemming

   ii. Removing non-alphanumeric characters

   (b) How do web search engines crawl the web? [*2 marks*]

   (c) What is a Permuterm Index? What is the main drawback of it? What should be the lookup query for "*tion" and for "blo*k"? [*2 marks*]

   (d) Politeness and Freshness are the main two considerations that web crawling should consider.

   i. How to balance between these two considerations? And why is that important? [*2 marks*]

   (e) If the following bit sequence is using v-encoding for a delta-encoded index, what is the corresponding delta-encoding for it? What is the original posting list (assuming the first posting stays the same)? [*1 mark*]

   10010010 00100011 00100100 10000001 00001001 10000101

2. A collection of 10,000 documents has been searched by systems S1 to S5 using query Q. Each system has returned the entire collection of 10,000 documents as a ranked list. The following table shows the top 20 documents returned by each system. Documents A, B, C, D, E, F, G and H are relevant. The letter N is used to denote the positions of non-relevant documents in the search results.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
S1	D	A	N	N	F	N	H	E	N	N	C	N	N	G	B	N	N	N	N	N
S2	N	E	N	H	A	C	N	N	N	N	D	N	F	N	N	N	N	N	N	N
S3	A	N	C	N	B	D	N	N	N	N	E	N	F	N	N	G	N	N	N	H
S4	E	C	N	A	N	D	N	B	N	N	N	N	H	N	N	N	N	F	G	N
S5	C	N	D	A	B	N	N	N	N	N	N	N	F	N	N	G	N	N	H	E

(a) Calculate each of the following metrics for each of the systems, then rank the systems by each of the metrics. Report your result in the format $S1 > S2 > S3 = S4 > S5$ where $S1 > S2$ means that system 1 performs better than system 2, and $S3 = S4$ means they perform the same. [*3 marks*]

i. Precision at R=50%

ii. R@6

iii. AP

(b) For each system, calculate the F1 score on the top 10 documents [*2 marks*]

(c) What is the largest possible MAP that the system S2 could have? Explain [*1 mark*]

(d) Assuming the graded relevance (gain) of documents are as follows: **A**=**B**=**3**, **C**=**D**=**E**=**2**, and **F**=**G**=**H**=**1**. What is the nDCG@10 for system S2. Show your steps. [*2 marks*]

(e) Assuming the same graded relevance of documents as above. Which system has the highest nDCG@1? Which one has the highest nDCG@2? [*2 marks*]

3. The query “brown fox” is run against the following collection of 8 documents:

d1: brown fox brown fox	d5: jump brown brown fox
d2: run watch fox jump	d6: brown brown brown jump
d3: fat brown fat fox	d7: watch jump run fox
d4: go fox brown go	d8: nice quite lovely fox

All statistics should be computed over this collection. Use base 10 for logarithms and do not add epsilon values.

(a) Which word (or words) have the highest idf value in this collection? Report the word (or words) and their idf value to three decimal digits. [*2 marks*]

(b) What is the Jaccard coefficient between d3 and d5? [*1 mark*]

(c) In the Language Modelling approach, what is the probability P(fat|d3). Assume Jelinek-Mercer smoothing with $\lambda = 0.5$. Report a number to three decimal digits. Show your work. [*2 marks*]

(d) Rank the documents according to the tf.idf weighted sum score for the query. Do not squash the tf component. Do not normalise for document length. Compute and report idf from the 8 documents (remember as always to show steps). Be sure to indicate ties. For example, if you think d1 would receive the highest score, followed by d2, d3 and d4 with the same score, followed by d5, then followed by d6-d8 all with the same score, then you would answer: [*3 marks*]

$$\text{d1} > \text{d2} = \text{d3} = \text{d4} > \text{d5} > \text{d6} = \text{d7} = \text{d8}$$

(e) The user indicates that d7 is relevant to the query, and d1 is non-relevant. Use the Rocchio algorithm to construct a new query. Set $\alpha = \beta = \gamma = 1$ and use tf.idf weights as in part (d) for all vectors. Do not normalise vectors to unit length. Report the resulting query vector. Show three decimal digits. Discuss the resulting query vector and compare it with the initial one. [*2 marks*]

4. You have collected a set of newspaper articles. Each article (document) belongs to exactly one category, such as *Cooking* or *Business*. You have identified a number of words that are of interest to you, such as *stocks* and *tree*, and counted how many documents in each category they appear in:

	Animals	Cooking	Tech	Arts	Business	Entertainment
Total no. of docs	2303	1361	803	1403	650	801
stocks	12	6	120	12	278	56
llama	452	24	320	41	3	3
biden	34	31	75	7	107	43
apple	56	156	310	23	2	41
alpaca	313	41	149	26	0	6
pear	4	167	0	3	5	107
tree	102	89	51	102	16	107

$$I(U;C) = \sum_{e_t \in \{1,0\}} \sum_{e_c \in \{1,0\}} P(U = e_t, C = e_c) \log_2 \frac{P(U = e_t, C = e_c)}{P(U = e_t)P(C = e_c)}$$

$$= \frac{N_{11}}{N} \log_2 \frac{NN_{11}}{N_{1.}N_{.1}} + \frac{N_{01}}{N} \log_2 \frac{NN_{01}}{N_{0.}N_{.1}} + \frac{N_{10}}{N} \log_2 \frac{NN_{10}}{N_{1.}N_{.0}} + \frac{N_{00}}{N} \log_2 \frac{NN_{00}}{N_{0.}N_{.0}}$$

(a) Calculate the expected mutual information $I(U;C)$ for the word *apple* and the category *Tech* (show steps).

(b) Calculate the expected mutual information $I(U;C)$ for *llama* and *Tech*.

(c) Which of the two words, *apple* and *llama*, occurs more often in documents of the category *Tech*? Which one has the higher expected mutual information? Briefly comment on your observation: if it is the same word for both questions, why do you think that is? If one word occurs more often but the other one has higher MI, why?

[*8 marks*]

5. (a) Describe the process of manual content analysis.

(b) Give an example of a situation where it makes sense to automate this process using a text classifier, and explain why.

(c) Give an example of a situation where it does not make sense to automate this process, and explain why. [*3 marks*]

6. Provide a short answer to each of the following questions:

(a) What is the difference between the unigram model and the mixture of unigrams model?

QUESTION CONTINUES ON NEXT PAGE

(b) Why is the generative process for LDA unrealistic? Use an example sentence to illustrate your point.

(c) i. Explain the term "out-of-vocabulary words" in the context of text classification.

ii. Name one approach we can use to deal with out-of-vocabulary words in traditional text classification (e.g. when training an SVM model from scratch).

iii. Explain briefly why this is less of an issue with transfer learning approaches.

(d) What is zero-shot classification and why is it useful? [*6 marks*]

7. Your colleague is building a search engine. She has put a great deal of effort into hand-crafting features, but she is not quite happy with the results yet. Name and describe three ways that query logs and click-through data might help her improve her search engine. [*3 marks*]